

Arpan Das

PhD Student and Graduate Research Assistant
Department of Soil and Crop Sciences
Texas A&M University, College Station, Texas, USA
arpan.das@tamu.edu | +19793448346
[Linkedin](#) | [Research gate](#) | [Orcid](#)
<https://arpandas.site>

SHORT BIO

Arpan Das is a plant breeder and geneticist specializing in wheat improvement for abiotic stress tolerance. His current research focuses on root system architecture and drought adaptation, using advanced phenotyping tools such as Magnetic Resonance Imaging (MRI). His goal is to advance climate-resilient crop breeding through the integration of modern phenotyping and genetic approaches.

ACADEMIC CREDENTIALS

Doctor of Philosophy in Plant Breeding (GPA: 4.0 out of 4 up to first year)

January, 2025 - Ongoing
Department of Soil and Crop Sciences
Texas A&M University, College Station, Texas, USA

Master of Science in Genetics and Plant Breeding (CGPA: 3.98 out of 4)

April, 2023 – November, 2024
Department of Genetics and Plant Breeding
Bangladesh Agricultural University, Mymensingh, Bangladesh

Bachelor of Science in Agriculture (CGPA: 3.786 out of 4)

January, 2018 – April, 2023
Faculty of Agriculture
Bangladesh Agricultural University, Mymensingh, Bangladesh

RESEARCH EXPERIENCE

• **Graduate Assistant – Research**

January, 2025 - Ongoing
Department of Soil and Crop Sciences
Texas A&M University, College Station, Texas, USA

Project 1: Developing drought-tolerant wheat through non-destructive root phenotyping of wild introgression lines

Responsibilities:

- Conducting wheat root phenotyping using advanced imaging technologies, including Magnetic Resonance Imaging (MRI).

- Integrating plant genetics and breeding approaches to investigate root system architecture and its role in adaptation to environmental stress.
- Collaborating with research team from Harvard University to develop and apply machine learning and AI methods for MRI-based wheat root image analysis.
- Screening of diverse wheat germplasm from Kansas State University to assess root system architecture traits for drought tolerance.
- Working with ABQMR Inc. to optimize MRI system parameters for high-quality wheat root scanning.

Project 2: Lab assessment of Biological Nitrification Inhibition (BNI) activity in wheat

Responsibilities:

- Established and managed hydroponic culture systems for controlled plant growth experiments.
- Collected and extracted root exudates from wheat for biochemical analysis.
- Screened wheat genotypes for Biological Nitrification Inhibition (BNI) activity.

- **Research Assistant (Plant Molecular Genetics Laboratory)**

April, 2023 – November, 2024

Department of Genetics and Plant Breeding

Bangladesh Agricultural University, Mymensingh, Bangladesh

Responsibilities:

- Conducted controlled crosses among diverse wheat and maize genotypes for hybrid development.
- Performed field-based morphological and molecular characterization of wheat genotypes for drought and terminal heat stress tolerance.
- Led breeding efforts in maize focusing on kernel color improvement and drought tolerance.
- Conducted field screening and evaluation of rice genotypes for high yield potential.
- Carried out morphological and biochemical assessments of key fruit quality traits in tomato.

RESEARCH INTEREST

- Molecular genetics and stress physiology
- Characterization for abiotic stress
- Advanced phenotyping
- Stress breeding
- Small grain quality improvement
- Molecular marker analysis
- Genome-wide association
- Bioinformatics and advanced phenomics

LANGUAGE SKILL

- Bengali (Native Speaker)
- English; IELTS overall Band 7.5 (L – 8, R – 8, S – 6.5, W – 6.5)
- Hindi (General communication)

SKILLS

- Advanced phenotyping using Magnetic Resonance Imaging (MRI)
- Command-line interface tools (MobaXterm, UNIX/Linux environment)
- DNA analysis, PCR based screening, and hybridization
- Proficient in general tools (Word, excel, PowerPoint, Photoshop)
- Data analysis (R, Minitab, STAR, SPSS) and interpretation of result
- Genome sequence editing (Mega, Bioedit and Genedoc)
- Photography (Participated in 2 exhibitions)
- Debate (Participated in Intra-hall and intra-varsity debate competitions)
- Content writing (written blog for many technology and travel websites)
- Graphics design (worked in digital content team and as freelance designer)
- Communication skills (Public speaking, leadership, program management, team work, critical thinking)

PUBLICATIONS

- Das, A., Biswas, A., Tandra, J. P., Beli, M. I., Islam, F. A., Niloy, T. A., ... & Sakil, M. A. (2026). Crop adaptation to climate change: Mechanisms, strategies, and future possibilities. *CABI Agriculture and Bioscience*, 7(1), 0067. <https://doi.org/10.1079/ab.2026.0067>
- **Das, A.**, Islam, M. A., Muntaha, S., Jahan, I., & Sagor, G. H. M. (2025). Combining Ability and Heterosis Analysis in Wheat (*Triticum aestivum* L.) Under Normal and Terminal Heat Stress Conditions. *International Journal of Agronomy*, 2025(1), 5656739. <https://doi.org/10.1155/ioa/5656739>
- Islam, M. A., **Das, A.**, Muntaha, S., & Sagor, G. H. M. (2026). Selection of High-yielding Genotypes and Phenotypic Correlations of F2 Wheat (*Triticum aestivum* L.) During the Late Planting Season in Field Conditions of Bangladesh. *Journal of Agricultural Sciences–Sri Lanka*, 21(2). <https://doi.org/10.4038/jas.v21i2.11191>
- Al Islam, F., Akter, A., Konica, S. I., Uddin, M. S., **Das, A.**, Begum, M., ... & Sakil, M. A. (2026). Soil Microbial Responses to Climate Change: From Mechanisms to Mitigation Strategies. *Cureus Journals*, 2(1). <https://doi.org/10.7759/s44497-025-00071-8>
- Angon, P. B., Mondal, S., Roy, A. R., **Das, A.**, Tithi, F. Y., Basak, A. R., ... & Sakil, M. A. CRISPR-driven genome modification and biofortification: Innovative techniques for creating climate-resistant and nutritionally improved crops. *Frontiers in Plant Physiology*, 4, 1900397. <https://doi.org/10.3389/fphgy.2026.1900397>
- Mondal, S., Angon, P. B., Uddin, M. S., **Das, A.**, Mondol, M., & Islam, M. S. Plant Genome Editing for Viral Disease Management: Current Achievements and Future Directions. In *Plant Stress Tolerance* (pp. 241-254). CRC Press. <http://dx.doi.org/10.1201/9781003543237-11>

- Angon, P.B., Mondal, S., **Das, A.**, Uddin, M.S., Eva, A.A. (2024). Micro-nanoplastics in the Environment: Current Research and Trends. In: Bhat, S.A., Kumar, V., Li, F., Kumar, S. (eds) Management of Micro and Nano-plastics in Soil and Biosolids. Springer, Cham. https://doi.org/10.1007/978-3-031-51967-3_5
- Angon, P. B., **Das, A.**, Roy, A. R., Khan, J. J., Ahmad, I., Biswas, A., ... & Yeasmin, S. T. (2024). Plant development and heat stress: role of exogenous nutrients and phytohormones in thermotolerance. *Discover Plants*, 1(1), 17. <https://doi.org/10.1007/s44372-024-00020-3>
- Chaki, T., **Das, A.**, Nikson, A. K., Muntaha, S. and Sagor, G. H. M. (2024). Genetic Analyses of Grain Yield and Yield-attributing Traits in Wheat (*Triticum aestivum* L.) Grown under Terminal Heat Stress. *Journal of Bangladesh Agricultural University*, 22(1), 81-91. <https://doi.org/10.5455/JBAU.184856>
- Das, A., **Das, A.**, Muntaha, S., Akhter, S., and Sagor, G. H. M. (2024). Genetic variability, correlation and path analysis of yield and yield contributing characters in wheat (*Triticum aestivum* L.) under normal and terminal heat stress condition. *Fundamental and Applied Agriculture*, 9(2), 65–71. <https://doi.org/10.5455/faa.187752>
- Angon, P. B., Anjum, N., **Das, A.**, Shreejana, K. C., Poudel, A., Suchi, S. A., & Islam, M. S. (2023). Sources, effects and present perspectives of heavy metals contamination: soil, plants and human food chain. *Heliyon*, 10(7). <https://doi.org/10.1016/j.heliyon.2024.e28357>
- Chowdhury, R., **Das, A.**, & Sagor, G. (2023). Characterization of Tomato Genotypes for Important Fruit Quality Traits. *European Journal of Theoretical and Applied Sciences*, 1(5), 267-280. [https://doi.org/10.59324/ejtas.2023.1\(5\).18](https://doi.org/10.59324/ejtas.2023.1(5).18)
- **Das, A.** (2023). Breeding History of Three Mega Rice (*Oryza sativa* L.) Varieties (BR11, BRRI dhan28, BRRI dhan29) of Bangladesh—A Review. <https://doi.org/10.32942/X2K02N>

CONFERENCE/POSTERS

- **Das, A.**, McFarland, A., Akhunov, E., Liu, S., & Okumoto, S. (2025). Root Phenotyping of Synthetic Hexaploidy Wheat Line for Drought-Tolerant Wheat. CANVAS 2025.
- Chowdhury, R., **Das, A.**, & Sagor, G. (2023). Characterization of Tomato Genotypes for Important Fruit Quality Traits

PRESENTATION AND MEETINGS

- CANVAS – American Society of Agronomy, Crop Science and Soil Science-Tri Society Annual Meeting 2025, Salt Lake City, Utah
- Plant Breeding and Genetics Circle, Dept. of Soil & Crop Sciences, Texas A&M University
- Wheat Genetics Resource Center Industry/University Cooperative Research Center (WGRC I/UCRC) Semi-Annual Meeting 2025, Manhattan, Kansas

- Small Grains Workers Meeting 2025, College Station, Texas

PROFESSIONAL ORGANIZATION

- National Association for Plant Breeding [2026 – present]
- Wheat Genetics Resource Center Industry/University Cooperative Research Center (WGRC I/UCRC) [2025 – present]
- American Society of Agronomy (ASA) [2025 – present]
- Crop Science Society of America (CSSA) [2025 – present]
- Soil Science Society of America (SSSA) [2025 – present]

TRAINING/WORKSHOP

- GRAD Aggies Lead Graduate Leadership Accelerator Program [Spring 2026]
- Graduate Mentoring Academy (GMA) Sessions, Texas A&M University [All 8 sessions attended]
- R Workshop for Data Analysis, Texas A&M Institute of Data Science, February, 2026
- Open Source for Open Science (OSOS), Texas A&M Institute of Data Science, October, 2025
- Data Analysis and Interpretation of Plant Breeding Research, Department of Genetics and Plant Breeding, Bangladesh Agricultural University, July, 2023
- Advanced Data Analysis Using Industry Accepted and Widely Popular Statistical Package, Bangladesh Research & Education Network (BdREN), Nov, 2020.
- MS Excel Basics of Beginners, Bohubrihi, May, 2020.

EXTRA-CURRICULUM / OTHER EXPERIENCES

- Social Chair, Texas A&M Plant Breeding Symposium Committee, Texas A&M University, College Station, Texas [2025 - present]
- Publication Secretary, Bangladesh Agricultural University Debating Shongho (BAUDS) [2021 – 2024]
- Digital Content Creator, Bangladesh Agricultural University Career Club (BAUCC) [2020 – 2022]
- Enzyme, Bangladesh Biology Olympiad (BdBO) [2019 – 2020]
- Photographer, Bangladesh Agricultural University Photographic Society (BAUPS) [2019 – 2021]
- Volunteer and Campus Ambassador, Aim Initiative Foundation (AIF) [2021 – 2022]

AWARDS

- G. Rollie White Fellowship for Resilient Agriculture Research (2026-27)
- GRAD Aggies Professional Development Foundation (2026)
- Graduate Mentoring Academy Fellow (2026)
- National Science and Technology (NST) Fellowship (2023)
- Winner of The MS hacks Campaign at 10 Minute School (2020)
- Finalist at Hult Prize at Bangladesh Agricultural University (2020)
- Champion in Bangladesh Physics Olympiad at Barisal Region (2015)

- Education Board Scholarship (General - SSC) (2015)
- Education Board Scholarship (Talentpool-PSC), 2nd in Barisal Board (2010)